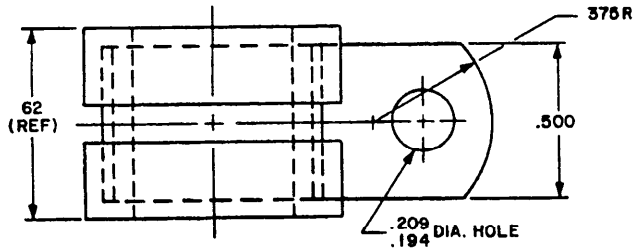
MIL-C-85052/2B
30 MAY 1984
SUPERSEDING
MIL-C-85052/2A
15 SEPTEMBER 1980

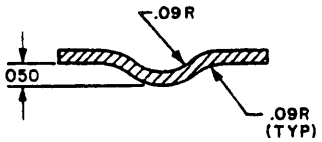
MILITARY SPECIFICATION SHEET

CLAMP, LOOP, TUBE-
17-7 PH CRES, 275° F, PHOSPHATE ESTER FLUID RESISTANT

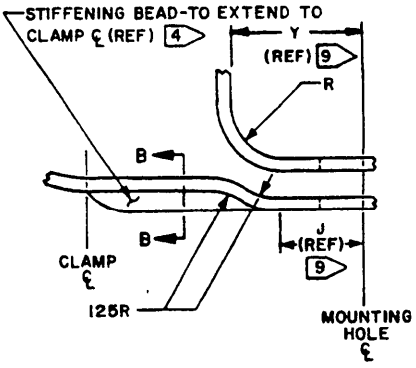
THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

THE COMPLETE REQUIREMENTS FOR ACQUIRING THE CLAMPS DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE LATEST ISSUE IN EFFECT OF SPECIFICATION MIL-C-85052.

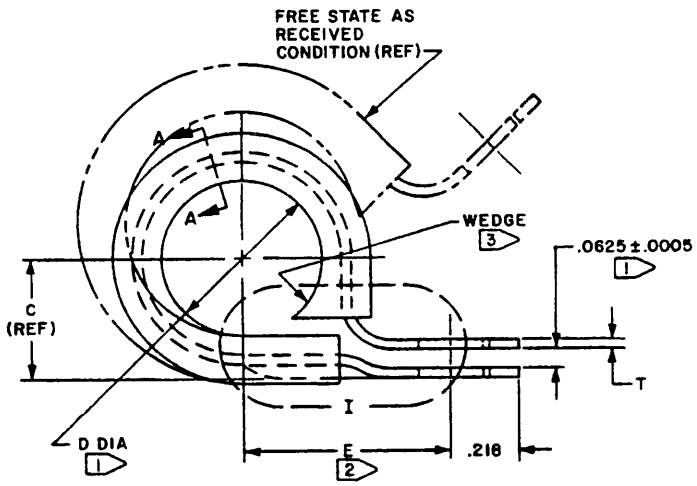




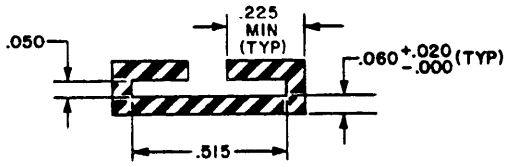
SECTION B-B



DETAIL I
(CUSHION REMOVED)



TOLERANCES: .XXX ± .010, .XX ± .03 UNLESS OTHERWISE SPECIFIED.



SECTION A-A
(CUSHION ONLY)

(B) DENOTES CHANGES

FSC 5340

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TABLE I

DASH NUMBER	C (REF)	D DIA [1]	E ± .032 [2]	J (REF)	R ±.010	T	Y (REF)
2	.192	.125	.468	--	.090	.020 ± 002	.325
3	224	.188	499	--			
4	.255	.250	530	.235			
5	.286	.312	561				
6	.318	375	592				
7	.349	438	624				
8	.380	.500	655				
9	.423	562	.741	.256	125	.032 ± 002	368
10	.454	.625	772				
11	486	688	804				
12	.517	750	835				
13	.548	812	866				

(B)

DASH NUMBER	C (REF)	D DIA. [1]	E ± 032 [2]	J (REF)	R ± 010	T	Y (REF)
14	580	875	898	256	125	.032 ± 002	368
15	611	938	929				
16	642	1 000	.960				
17	681	1.062	1 001	262		.040 ± 003	.370
18	712	1 125	1 032				
19	744	1 188	1 064				
20	775	1 250	1 095				
21	806	1 312	1 126				
22	838	1 375	1 158				
23	869	1 438	1 189				
24	900	1 500	1 220				

(B)

NOTES

[1] DIAMETER "D" IS THE NOMINAL DIAMETER FOR WHICH A CLAMP SIZE IS INTENDED FOR USE. DIAMETER "D" SHALL BE VERIFIED BY THE "DIAMETRAL RETENTION" TEST OF THE GENERAL SPECIFICATION

[2] DIMENSION "E" SHALL BE MEASURED WITH THE CLAMP INSTALLED ON A MANDREL OF "D" DIAMETER ± 001 AND AN 0625 ±.0005 SPACER BETWEEN THE UPPER AND LOWER FOOT AS SHOWN

(B) [3] WEDGE SHALL BE REQUIRED ON SIZE -6 AND LARGER. THE WEDGE SHALL BE MOLDED AS AN INTEGRAL PART OF THE CUSHION AND CONTOURED TO FIT "D" DIAMETER

(B) [4] STIFFENING BEAD SHALL BE REQUIRED ON SIZE -4 AND LARGER. ALL RADII ON THE STIFFENING BEAD SHALL BE SMOOTH AND BLENDED, NO SHARP TOOL MARKS ARE ALLOWED

(B) 5. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE INCHES

6. METAL BANDS SHALL HAVE ALL BURRS, SHARP EDGES AND SCALE REMOVED

(B) [7] WEDGE SHALL TOUCH CUSHION ON LOWER FOOT WITH CLAMP INSTALLED ON MANDREL WITHOUT SPACER

(B) [8] IF MIL-C-85052/4 CLAMP SUPPORTS ARE USED, THE MAXIMUM TUBE OVERHANG SHALL BE 1/2 TUBE O.D. THIS WASHER CAN BE DETRIMENTAL IF THE CLAMP IS NOT FULLY SUPPORTED

[9] REFERENCE DIMENSIONS "Y" AND "J" ARE PROVIDED TO GAIN MAXIMUM SUPPORT FOR THE LOWER FOOT BY CLOSELY FITTING THE UPPER FOOT BEND RADIUS TO THE STIFFENING BEAD BLEND RADIUS WHEN CLOSED AGAINST EACH OTHER

[10] UNLESS OTHERWISE SPECIFIED, PROPERTY MEASUREMENTS SHALL BE PERFORMED IMMEDIATELY AFTER REMOVAL FROM THE IMMERSION BATH

REQUIREMENTS

MARKING

BAND - THE COMPLETE STANDARD PART NUMBER AND MANUFACTURER'S NAME, TRADEMARK, OR FSCM NUMBER SHALL BE IMPRESSION STAMPED ON THE BAND IN AN AREA NOT COVERED BY THE CUSHION. ALL MARKING SHALL BE ABOVE TUBE THEORETICAL CENTERLINE

- CLAMP BANDS OF -2, -3, -4 SIZES MAY BE MARKED WITH THE MANUFACTURERS INDENTIFICATION, THE SIZE, AND THE SPECIFICATION SHEET NUMBER DUE TO SPACE LIMITATIONS

CUSHION - NONE

MATERIALS.

METAL BAND - 17-7PH, CORROSION RESISTANT STEEL PER MIL-S-25043, ANNEALED, STRESS RELIEVE AND HEAT TREAT TO TH1100 CONDITION PER MIL-H-6875, AFTER FORMING.

CUSHION - ETHYLENE PROPYLENE RUBBER, 65-75 DUROMETER, COLOR - PURPLE, PER THE GENERAL SPECIFICATION WITH ADDITIONAL REQUIREMENTS SHOWN ON THIS SPECIFITION SHEET

FINISH

METAL BAND - PASSIVATE PER QQ-P-35

CUSHION - NONE

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PART NUMBER

PART NUMBERS SHALL CONSIST OF THE FOLLOWING (IN SEQUENCE)

1 THE LETTER "M"
2 THE GENERAL SPECIFICATION NUMBER
3 A SLASH AND SLASH NUMBER OF THIS SPECIFICATION SHEET
4 A DASH, AND THE APPROPRIATE SIZE DASH NUMBERS FROM TABLE I

EXAMPLE M85052/2-8

CUSHION REQUIREMENTS

1 PHYSICAL PROPERTIES - AS SPECIFIED IN TABLE II UNLESS OTHERWISE SPECIFIED RESULTS ARE AN AVERAGE OF 5 SPECIMENS AND TOLERANCE ON TEMPERATURE = ± 5°F

TABLE II

TEST	TEST METHOD	REQUIRED ORIGINAL PROPERTIES	ALLOWABLE CHANGE FROM ACTUAL ORIGINAL PROPERTIES AFTER	
			HYDRAULIC FLUID IMMERSION SKYDROL 500B 22 HRS AT 275°F (10)	HEAT AGING 70 HRS AT 275°F
HARDNESS, DUROMETER "A"	ASTM D2240	65-75	-20 MAX	+10 MAX
TENSILE STRENGTH - PSI	ASTM D412	2000 MIN	-50% MAX	-20% MAX
ELONGATION - %	ASTM D412	400 MIN	-25% MAX	-40% MAX
TEAR STRENGTH - PPI	ASTM D624 DIE "B"	200 MIN	-40% MAX	-40% MAX
SPECIFIC GRAVITY	ASTM D297	REPORT	--------	+2%
VOLUME CHANGE - %	ASTM D471-75	--------	+30% MAX	-10% MAX

(B) 2 COMPRESSION SET - SEE MIL-C-85052
- AIR AGE AT 212 F
- NOT TO EXCEED 30% AVERAGE OF 3 SPECIMENS

(B) 3 FLAMMABILITY - SEE MIL-C-85052
- SPECIMENS - STANDARD ASTM TEST SLABS
- HORIZONTAL BURN TEST

INTENDED USE

THESE CLAMPS ARE INTENDED FOR USE AS FOLLOWS

TEMPERATURE RANGE

-65°F TO +275°F (FOR HIGHER TEMPERATURE RATED CLAMPS SEE MIL-C-85052/3 CLAMPS)

SYSTEMS

ALL FLUID AND ELECTRICAL SYSTEMS ON AIRCRAFT AND AEROSPACE VEHICLES USING PHOSPHATE ESTER BASED HYDRAULIC SYSTEMS, EXCEPT APPLICATIONS REQUIRING A HIGH DEGREE OF FUEL EXPOSURE USE MIL-C-85052/1 CLAMPS

VIBRATION RATING

DEPENDING ON MOUNTING METHODS (SEE RECOMMENDED MOUNTING DATA)

Custodians
NAVY - AS
AIR FORCE -99
ARMY - AV

Review Activities
AIR FORCE - 11
DLA - IS

Preparing Activity
NAVY - AS

PROJECT NO 5340-1573

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RECOMMENDED MOUNTING METHODS

MOUNTING METHODS AFFECT THE VIBRATION CAPABILITIES OF THESE CLAMPS. TYPICAL MOUNTING METHODS AND VIBRATION RATINGS ARE SHOWN IN FIGURES I, II, AND III THE METHOD SHOWN IN FIGURE II IS RECOMMENDED WHENEVER POSSIBLE. THE METHOD SHOWN IN FIGURE III SHOULD ONLY BE USED IN VERY HIGH VIBRATION AREAS OR AS NECESSARY.

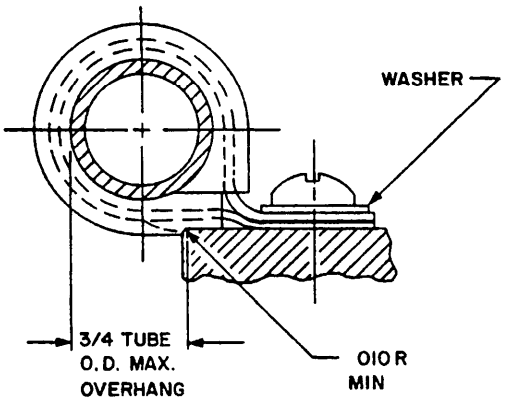


FIGURE I
LOW TO MEDIUM VIBRATION AREAS

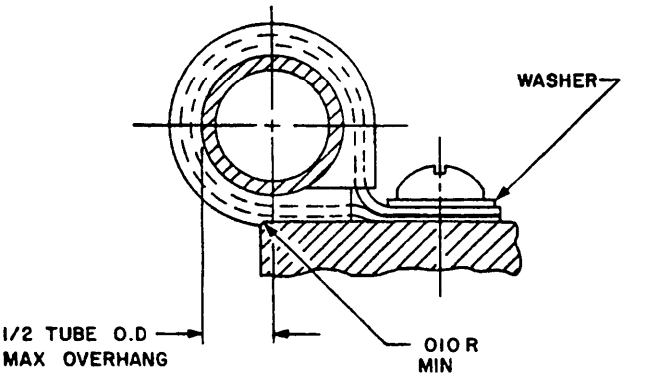


FIGURE II
LOW TO HIGH VIBRATION AREAS

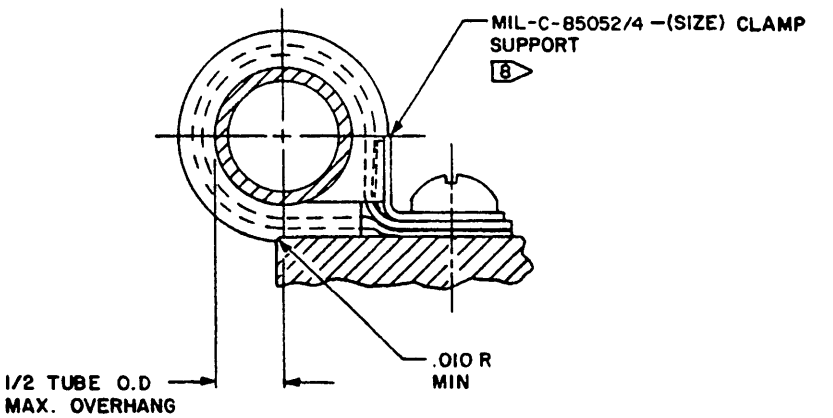


FIGURE III
VERY HIGH VIBRATION AREAS

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INCH–POUND

NOTICE
OF VALIDATION

MIL–C–85052/2B
NOTICE 1
26 September 1991

MILITARY SPECIFICATION

CLAMP, LOOP, TUBE 17–7 PH CRES, 275 DEG. F, PHOSPHATE ESTER FLUID RESISTANT

MIL–C–85052/2B, dated 30 May 1984, has been reviewed and determined to be valid for use in acquisition.

Custodians:
Army – AV
Navy – AS
Air Force – 99

Preparing activity:
Navy – AS

AMSC N/A

FSC 5340

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